OPERATING SYSTEMS – ASSIGNMENT 1

TASK 9 – Evaluate the Memory Hierarchy and I/O Systems (4 Marks)
CO1–CO5 | Bloom's Level: K5 – Evaluate

Introduction

A computer uses different levels of memory and storage. Faster levels are usually smaller and more expensive per byte, while slower levels provide larger storage capacity. The hospital must place data at the correct level and select a suitable I/O method.

1. Memory hierarchy

Registers: Located inside the CPU; extremely fast and very small. They hold values currently being processed.
Cache: Very fast memory close to the CPU; stores frequently used instructions and data.
Main memory (RAM): Holds active programs and data currently in use. It is larger than cache but slower.
SSD: Non-volatile storage with fast access; suitable for operating systems, databases and active medical files.
Hard disk: Large-capacity storage, usually slower than SSD; suitable for archives and less frequently accessed data.
Network storage: Allows data to be stored and shared across servers or locations; access time depends on the network and storage system.

2. Suitable storage for hospital data

MRI images: Store active images on fast SSD or high-performance storage for processing. Move older images to larger archive or network storage according to retention policy.
Patient records: Store in a reliable database on redundant server storage with backups and access controls. Frequently accessed records may be cached in RAM, but the database remains the authoritative copy.
Monitor readings: Keep recent readings in memory or local storage for quick access, and transmit required records securely to central storage. Important readings should not be lost if a network connection is interrupted.

3. Programmed I/O, interrupt-driven I/O and DMA

Programmed I/O: The CPU repeatedly checks device status and moves data. It is simple but can waste CPU time.
Interrupt-driven I/O: The device interrupts the CPU when it needs attention or has completed an operation. It avoids continuous checking.
Direct Memory Access (DMA): A DMA controller transfers blocks of data between a device and main memory with limited CPU involvement. The CPU is typically notified when the transfer completes.

4. Suitable I/O method for MRI scanner

DMA is suitable for an MRI scanner because scan data can be large. It allows blocks of image data to move into memory without requiring the CPU to handle every word. Interrupt-driven notification can be used when the transfer completes. This frees CPU time for image processing and other tasks.

5. Effect of caching on record lookup

A cache stores frequently accessed records or database pages in faster memory. If the requested data is in the cache (cache hit), the system can return it faster without accessing slower storage. If it is not present (cache miss), the system must fetch it from the database storage. Cache design must maintain consistency and protect confidential data.

6. Alternative concept: one interrupt per word for a 500 MB MRI image

If interrupt-driven I/O generated one interrupt for every word, the number of interrupts would be extremely large. Assuming a 4-byte word and 500 MB = 500 × 1,000,000 bytes, the transfer would involve approximately 125,000,000 word interrupts. If 500 MiB is intended, the count would be 131,072,000 interrupts. Handling each interrupt consumes CPU time and creates overhead, which can delay other work. With DMA, the device transfers a block of data and normally generates far fewer completion notifications. Both approaches need memory buffers; DMA typically uses one or more block buffers, while interrupt-per-word handling still needs a receiving buffer and many CPU service operations.

Conclusion

The hospital should use fast memory and SSD storage for active work, reliable database storage for patient records, archive storage for older images and DMA for large MRI data transfers.

Prepared from the corresponding task in the uploaded CSA04 Operating Systems Assignment 1 document. Add your name, register number, required screenshots and your own measured evidence wherever requested before submission.